The Revolutionary Five-Body Problem

Hierarchical Celestial Mechanics, the Earth–Moon Bloc, Strategic Compression, and the Emergence of Four Effective Units from Five Physical Bodies

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Abstract

This paper introduces the *Revolutionary Five-Body Problem*, a hierarchical extension of the Strategic Four-Body Problem in which the five physical bodies are the Sun, Mercury, Venus, Earth, and Moon. The fifth body does not enter the system symmetrically: the Moon is gravitationally bound to the Earth while the Earth–Moon barycenter participates in the larger heliocentric architecture. This creates a natural distinction between the number of physical bodies and the number of effectively independent dynamical or strategic units.

The system is written as

$$\mathcal{B} = \{\mathrm{S}, \mathrm{Me}, \mathrm{V}, \mathrm{E}, \mathrm{L}\},$$

while a reduced hierarchical representation is

$$\mathcal{U} = \{\mathrm{S}, \mathrm{Me}, \mathrm{V}, \mathrm{EM}\}.$$

Thus

$$N_B = 5, \qquad N_{\mathrm{eff}} = 4.$$

The Earth–Moon subsystem is decomposed exactly into barycentric and internal coordinates. At leading order, the full problem becomes a hierarchy of four Keplerian structures: Sun–Mercury, Sun–Venus, Sun–Earth–Moon barycenter, and Earth–Moon relative motion. Planetary perturbations, solar tides, relativistic corrections, multipole effects, and other interactions can then be incorporated systematically.

The resulting architecture supplies a concrete test of the distinction between physical multiplicity and informational or strategic independence. A fifth physical body need not constitute a fifth independent strategic unit when hierarchical binding, coalition formation, barycentric compression, common information, or multiscale organization permits a composite subsystem to act externally as a single effective unit while retaining internal degrees of freedom.

The paper develops this idea using celestial mechanics, astronomy, astrophysics, cosmology, physics, information theory, statistical mechanics, economics, finance, political science, international relations, welfare economics, strategic conflict, psychology, philosophy, statistics, computer science, data science, artificial intelligence, machine learning, network science, control theory, and hierarchical systems theory. The principal result is not an exact closed-form solution of the generic five-body problem, but a mathematically explicit program for solving this particular hierarchical system through decomposition, perturbation, observation, inference, compression, and recursively improvable approximation.

The paper ends with “The End”

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1 Introduction

The classical five-body problem concerns five masses interacting through gravity. For bodies $i = 1, \ldots, 5$,

$$m_i \ddot{\boldsymbol{r}}_i = -G m_i \sum_{\substack{j=1 \\ j \neq i}}^{5} m_j \frac{\boldsymbol{r}_i - \boldsymbol{r}_j}{|\boldsymbol{r}_i - \boldsymbol{r}_j|^3}.$$

A generic five-body system is substantially more complicated than the two-body Kepler problem. The present paper does not attack the completely generic case.

Instead, we choose a physically meaningful hierarchy:

$$\boxed{B_1 = \text{Sun}, \quad B_2 = \text{Mercury}, \quad B_3 = \text{Venus}, \quad B_4 = \text{Earth}, \quad B_5 = \text{Moon}.}$$

The key observation is that the fifth body is not simply another heliocentric planet. The Moon is a satellite participating in an internal Earth–Moon subsystem.

Consequently, the architecture is approximately

$$\boxed{\text{Sun} \longrightarrow \begin{cases} \text{Mercury}, \\ \text{Venus}, \\ (\text{Earth}, \text{Moon}). \end{cases}}$$

This hierarchy suggests a general principle:

> The number of physical bodies need not equal the number of effectively independent dynamical, informational, political, or strategic units.

We call the resulting problem the *Revolutionary Five-Body Problem*. The terminology is deliberately double-sided: celestial bodies revolve, but the addition of a hierarchically bound fifth body also changes the conceptual architecture of the preceding four-body framework.

2 From the Strategic Four-Body Problem to Five Bodies

The Strategic Four-Body Problem considered the architecture

$$\{\mathrm{S}, \mathrm{Me}, \mathrm{V}, \mathrm{E}\}.$$

Its physical interpretation already possesses a useful hierarchy because

$$M_{\mathrm{S}} \gg m_{\mathrm{Me}}, m_{\mathrm{V}}, m_{\mathrm{E}}.$$

At leading order, the system can therefore be represented by three Kepler problems around a dominant central mass.

The introduction of the Moon gives

$$\{\mathrm{S}, \mathrm{Me}, \mathrm{V}, \mathrm{E}, \mathrm{L}\}.$$

But the Moon introduces a qualitatively different relationship:

$$\mathrm{L} \leftrightarrow \mathrm{E}.$$

The resulting hierarchy contains both heliocentric and geocentric/barycentric scales.

This is the source of the proposed compression.

3 Physical Five-Body System

Define

$$\mathcal{B} = \{\mathrm{S}, \mathrm{Me}, \mathrm{V}, \mathrm{E}, \mathrm{L}\}.$$

Each body has state

$$X_i = (\boldsymbol{r}_i, \boldsymbol{v}_i, m_i).$$

The complete physical state is

$$X = (X_{\mathrm{S}}, X_{\mathrm{Me}}, X_{\mathrm{V}}, X_{\mathrm{E}}, X_{\mathrm{L}}).$$

The Newtonian Hamiltonian is

$$H = \sum_{i=1}^{5} \frac{\boldsymbol{p}_i^2}{2m_i} - G \sum_{1 \le i < j \le 5} \frac{m_i m_j}{r_{ij}}.$$

The complete graph contains

$$\binom{5}{2} = 10$$

pairwise gravitational interactions.
The phase-space dimension is

$$6N_B = 30.$$

4 Relative Configuration Dimension

Five bodies possess

$$3N_B = 15$$

position coordinates.
Removing the three coordinates associated with overall translation gives

$$15 - 3 = 12.$$

Theorem 4.1 (Five-Body Relative Dimension)**.** *The translationally reduced configuration space of five point bodies in three-dimensional Euclidean space has dimension*

$$\boxed{\dim \mathcal{Q}_{\mathrm{rel}}^{(5)} = 12.}$$

Proof. There are five position vectors in $\mathbb{R}^3$, hence fifteen scalar position coordinates. The collective center-of-mass position contributes three coordinates. Removing them leaves

$$15 - 3 = 12.$$

Therefore

$$\dim \mathcal{Q}_{\mathrm{rel}}^{(5)} = 12.$$

□

This is immediately different from the four-body result

$$3(4) - 3 = 9.$$

Thus the nine-dimensional coincidence of the Strategic Four-Body Problem does not extend trivially to five physical bodies.

5 Earth–Moon Barycentric Decomposition

Define the Earth–Moon mass

$$M_{\mathrm{EM}} = m_{\mathrm{E}} + m_{\mathrm{L}}.$$

Define their barycenter

$$\boxed{\boldsymbol{R}_{\mathrm{EM}} = \frac{m_{\mathrm{E}}\boldsymbol{r}_{\mathrm{E}} + m_{\mathrm{L}}\boldsymbol{r}_{\mathrm{L}}}{M_{\mathrm{EM}}}.}$$

Define the internal Earth–Moon relative coordinate

$$\boxed{\boldsymbol{\rho}_{\mathrm{EM}} = \boldsymbol{r}_{\mathrm{L}} - \boldsymbol{r}_{\mathrm{E}}.}$$

These variables permit exact reconstruction.

Theorem 5.1 (Earth–Moon Reconstruction)**.** *The Earth and Moon positions are exactly recovered from* $\boldsymbol{R}_{\mathrm{EM}}$ *and* $\boldsymbol{\rho}_{\mathrm{EM}}$ *according to*

$$\boxed{\boldsymbol{r}_{\mathrm{E}} = \boldsymbol{R}_{\mathrm{EM}} - \frac{m_{\mathrm{L}}}{M_{\mathrm{EM}}}\boldsymbol{\rho}_{\mathrm{EM}}}$$

and

$$\boxed{\boldsymbol{r}_{\mathrm{L}} = \boldsymbol{R}_{\mathrm{EM}} + \frac{m_{\mathrm{E}}}{M_{\mathrm{EM}}}\boldsymbol{\rho}_{\mathrm{EM}}.}$$

Proof. From

$$\boldsymbol{\rho}_{\mathrm{EM}} = \boldsymbol{r}_{\mathrm{L}} - \boldsymbol{r}_{\mathrm{E}},$$

we have

$$\boldsymbol{r}_{\mathrm{L}} = \boldsymbol{r}_{\mathrm{E}} + \boldsymbol{\rho}_{\mathrm{EM}}.$$

Substitution into the barycenter equation gives

$$\boldsymbol{R}_{\mathrm{EM}} = \frac{m_{\mathrm{E}}\boldsymbol{r}_{\mathrm{E}} + m_{\mathrm{L}}(\boldsymbol{r}_{\mathrm{E}} + \boldsymbol{\rho}_{\mathrm{EM}})}{M_{\mathrm{EM}}}.$$

Therefore

$$\boldsymbol{R}_{\mathrm{EM}} = \boldsymbol{r}_{\mathrm{E}} + \frac{m_{\mathrm{L}}}{M_{\mathrm{EM}}}\boldsymbol{\rho}_{\mathrm{EM}},$$

which implies

$$\boldsymbol{r}_{\mathrm{E}} = \boldsymbol{R}_{\mathrm{EM}} - \frac{m_{\mathrm{L}}}{M_{\mathrm{EM}}}\boldsymbol{\rho}_{\mathrm{EM}}.$$

Adding $\boldsymbol{\rho}_{\mathrm{EM}}$ gives the Moon equation. □

This transformation is exact. The approximation enters only when the external gravitational field is subsequently expanded around the Earth–Moon barycenter.

6 Earth–Moon Kinetic-Energy Separation

Define the reduced mass

$$\mu_{\mathrm{EM}} = \frac{m_{\mathrm{E}} m_{\mathrm{L}}}{M_{\mathrm{EM}}}.$$

Then

$$\frac{1}{2}m_{\mathrm{E}}\dot{\boldsymbol{r}}_{\mathrm{E}}^2 + \frac{1}{2}m_{\mathrm{L}}\dot{\boldsymbol{r}}_{\mathrm{L}}^2 = \frac{1}{2}M_{\mathrm{EM}}\dot{\boldsymbol{R}}_{\mathrm{EM}}^2 + \frac{1}{2}\mu_{\mathrm{EM}}\dot{\boldsymbol{\rho}}_{\mathrm{EM}}^2.$$

Thus external bloc motion and internal relative motion separate exactly in the kinetic energy.
This is the mechanical foundation of the effective-unit interpretation.

7 Five Bodies but Four Effective External Units

Define the physical body count

$$N_B = 5.$$

Now define the set of effective external units

$$\boxed{\mathcal{U} = \{\mathrm{S}, \mathrm{Me}, \mathrm{V}, \mathrm{EM}\}.}$$

Hence

$$\boxed{N_{\text{eff}} = 4.}$$

The fifth physical body survives as an internal degree of freedom of the Earth–Moon composite.

Definition 7.1 (Effective Strategic Unit)**.** *An effective strategic unit is a body or bound subsystem whose external state can, over a specified scale and accuracy, be represented by a common macroscopic state while its internal degrees of freedom are modeled separately.*

The central distinction is therefore

$$\boxed{N_B \neq N_{\text{eff}}}$$

in general.

8 Hierarchical Compression Theorem

Theorem 8.1 (Barycentric Hierarchical Compression)**.** *Consider two bodies A and B embedded in a larger gravitational system. Their six positional coordinates can be represented exactly by the three coordinates of their barycenter and the three coordinates of their internal relative displacement:*

$$(\boldsymbol{r}_A, \boldsymbol{r}_B) \longleftrightarrow (\boldsymbol{R}_{AB}, \boldsymbol{\rho}_{AB}).$$

Consequently, external and internal descriptions can be separated without loss of positional information.

Proof. The forward transformation is

$$\boldsymbol{R}_{AB} = \frac{m_A \boldsymbol{r}_A + m_B \boldsymbol{r}_B}{m_A + m_B},$$

$$\boldsymbol{\rho}_{AB} = \boldsymbol{r}_B - \boldsymbol{r}_A.$$

The inverse transformation is

$$\boldsymbol{r}_A = \boldsymbol{R}_{AB} - \frac{m_B}{m_A + m_B} \boldsymbol{\rho}_{AB},$$

$$\boldsymbol{r}_B = \boldsymbol{R}_{AB} + \frac{m_A}{m_A + m_B} \boldsymbol{\rho}_{AB}.$$

Hence the transformation is bijective for positive finite masses. □

Remark 8.2. *This theorem does not reduce the exact number of physical degrees of freedom. It reorganizes them into external and internal sectors. Effective dimensional reduction occurs only when the internal sector can be averaged, compressed, marginalized, or otherwise treated at lower resolution.*

9 The Four Leading Kepler Problems

The hierarchy suggests the leading decomposition

$$H_0 = H^K_{\mathrm{SMe}} + H^K_{\mathrm{SV}} + H^K_{\mathrm{SEM}} + H^K_{\mathrm{EL}}.$$

Each Kepler Hamiltonian has the generic form

$$H^K_{AB} = \frac{\boldsymbol{p}^2_{AB}}{2\mu_{AB}} - \frac{G m_A m_B}{r_{AB}},$$

where

$$\mu_{AB} = \frac{m_A m_B}{m_A + m_B}.$$

The four leading orbital structures are therefore

$$\boxed{\begin{array}{l} \mathrm{S} \leftrightarrow \mathrm{Me}, \\ \mathrm{S} \leftrightarrow \mathrm{V}, \\ \mathrm{S} \leftrightarrow \mathrm{EM}, \\ \mathrm{E} \leftrightarrow \mathrm{L}. \end{array}}$$

Each isolated term is integrable.
The difficult dynamics are moved into the couplings among these subsystems.

10 Perturbative Hierarchy

Write

$$\boxed{H = H_0 + \epsilon H_1 + \epsilon^2 H_2 + \cdots .}$$

Here H_0 contains the leading Keplerian architecture.
The first perturbative level may contain

$$H_1 = H_{\text{planetary}} + H_{\text{solar tidal}} + H_{\text{lunar coupling}} + H_{\mathrm{GR}} + H_{\text{multipole}}.$$

Higher levels can incorporate smaller effects.
The approximation sequence is

$$X^{(0)}(t), \quad X^{(1)}(t), \quad X^{(2)}(t), \quad \ldots$$

where

$$X^{(0)} = \text{hierarchical Keplerian solution.}$$

The objective is not merely to generate approximations but to obtain controlled errors

$$\boxed{\|X(t) - X^{(n)}(t)\| \le \varepsilon_n(t).}$$

A recursively improvable solution with quantified error is a meaningful notion of solution for a hierarchical many-body problem.

11 Why the Hierarchy Works

The hierarchy is supported by two small-parameter structures.
First,

$$M_{\mathrm{S}} \gg m_{\mathrm{Me}}, m_{\mathrm{V}}, m_{\mathrm{E}}, m_{\mathrm{L}}.$$

Second, the Earth–Moon separation is much smaller than the heliocentric Earth–Moon distance:

$$a_{\mathrm{EL}} \ll a_{\mathrm{SEM}}.$$

Define

$$\epsilon_M = \frac{m_{\mathrm{planet}}}{M_{\mathrm{S}}}$$

schematically for planetary mass ratios and

$$\epsilon_L = \frac{a_{\mathrm{EL}}}{a_{\mathrm{SEM}}}.$$

The perturbative structure can then be organized by powers and combinations of such dimensionless quantities.

12 Solar Tidal Expansion of the Earth–Moon Subsystem

Let

$$\boldsymbol{R} = \boldsymbol{R}_{\mathrm{EM}} - \boldsymbol{r}_{\mathrm{S}}$$

be the Sun-to-Earth–Moon-barycenter vector and let

$$|\boldsymbol{\rho}_{\mathrm{EM}}| \ll |\boldsymbol{R}|.$$

The solar potential acting on Earth and Moon can be expanded around $\boldsymbol{R}$.

For a displacement $\boldsymbol{\delta}$,

$$\frac{1}{|\boldsymbol{R}+\boldsymbol{\delta}|} = \frac{1}{R} - \frac{\boldsymbol{R}\cdot\boldsymbol{\delta}}{R^3} + \frac{3(\boldsymbol{R}\cdot\boldsymbol{\delta})^2 - R^2\delta^2}{2R^5} + \cdots .$$

Because barycentric coordinates eliminate the mass-weighted dipole term, the leading internal correction is tidal/quadrupolar.

Thus the Sun sees the Earth–Moon pair approximately as a single mass M_{EM} at large distances, while internal lunar structure enters through higher-order multipoles.

This gives a precise mathematical meaning to the statement that Earth and Moon form one effective external unit.

13 Multipole Interpretation

At distances R much larger than the subsystem size ρ, the potential of a compact subsystem has schematic expansion

$$\Phi(R) = -\frac{GM}{R} + \Phi_{\mathrm{dipole}} + \Phi_{\mathrm{quadrupole}} + \cdots .$$

In the center-of-mass frame,

$$\Phi_{\mathrm{dipole}} = 0.$$

Therefore

$$\boxed{\Phi(R) = -\frac{GM}{R} + O\left(\frac{\rho^2}{R^3}\right).}$$

The monopole term depends only upon total mass.

Internal structure becomes visible at higher orders.

This is one of the strongest physical foundations for hierarchical strategic compression.

14 Fast and Slow Time Scales

The Moon introduces a second important orbital scale.

Let

$$\theta_L = n_L t + \phi_L$$

denote a fast lunar phase.

Let

$$\theta_E = n_E t + \phi_E$$

denote the slower annual phase.

More generally,

$$\theta_i = n_i t + \phi_i.$$

The hierarchy

$$n_L > n_E$$

motivates multiple-time-scale methods.

Introduce

$$t_0 = t, \qquad t_1 = \epsilon t, \qquad t_2 = \epsilon^2 t.$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}t} = \frac{\partial}{\partial t_0} + \epsilon \frac{\partial}{\partial t_1} + \epsilon^2 \frac{\partial}{\partial t_2} + \cdots .$$

Fast oscillations can be averaged while slow secular evolution is retained.

15 Secular Dynamics

Suppose the Hamiltonian is

$$H(J,\theta) = H_0(J) + \epsilon H_1(J,\theta).$$

Define the angle average

$$\overline{H}_1(J) = \frac{1}{(2\pi)^k} \int_{[0,2\pi]^k} H_1(J,\theta)\,\mathrm{d}^k\theta.$$

Then the secular Hamiltonian is approximately

$$H_{\text{sec}} = H_0 + \epsilon \overline{H}_1.$$

The resulting slow equations are

$$\dot{J}_i = -\frac{\partial H_{\text{sec}}}{\partial \theta_i},$$

$$\dot{\theta}_i = \frac{\partial H_{\text{sec}}}{\partial J_i}.$$

This provides a second form of compression: fast orbital detail is averaged to obtain slower macroscopic evolution.

16 Relativistic Corrections

Mercury provides an especially important test because relativistic effects are measurable in its orbit.

At leading post-Newtonian order, the relativistic perihelion advance per orbit is

$$\boxed{\Delta\varpi = \frac{6\pi G M_{\mathrm{S}}}{a(1-e^2)c^2}.}$$

Thus a high-precision model should have the schematic hierarchy

$$H = H_{\mathrm{Kepler}} + H_{\mathrm{planetary}} + H_{\mathrm{lunar}} + H_{\mathrm{1PN}} + H_{\mathrm{higher}}.$$

The Revolutionary Five-Body Problem is therefore naturally a Newtonian-plus-relativistic perturbation problem rather than a purely Newtonian construction when observational precision is sufficiently high.

17 Astronomical Observations

The system is unusually attractive because all five bodies are observationally accessible.

Let the latent state be X_t and observations satisfy

$$Y_t = h(X_t) + \epsilon_t.$$

Possible observations include

- angular position,
- ranging,
- radar measurements,
- Doppler velocity,
- eclipse and transit timing,
- lunar laser ranging,
- spacecraft tracking.

The posterior is

$$p(X_t \mid Y_{1:t}) \propto p(Y_t \mid X_t) p(X_t \mid Y_{1:t-1}).$$

Thus theoretical approximation and empirical observation can be updated recursively.

18 Strategic Astronomy

The observational loop is

$$\boxed{\text{Dynamics} \to \text{Observation} \to \text{Inference} \to \text{Model Revision} \to \text{Prediction} \to \text{Observation.}}$$

If the bodies are passive astronomical objects, the strategic layer belongs to the observers studying them.

If some bodies contain technological agents capable of maneuver, the loop can instead become

$$\boxed{\text{Dynamics} \to \text{Observation} \to \text{Inference} \to \text{Decision} \to \text{Control} \to \text{Dynamics.}}$$

These two interpretations must not be conflated.

19 Physical Bodies versus Strategic Units

We now distinguish three counts.

Let

$$N_B = \text{number of physical bodies},$$

$$N_D = \text{number of dynamically resolved subsystems},$$

and

$$N_{\text{eff}} = \text{number of effective strategic units}.$$

For the present construction,

$$N_B = 5.$$

At one external level,

$$N_{\text{eff}} = 4.$$

At finer resolution, Earth and Moon separate again.

Thus N_{eff} is scale-dependent.

Definition 19.1 (Resolution-Dependent Unit Count). *Let ℓ denote observational or dynamical resolution. Define*

$$N_{\text{eff}}(\ell)$$

as the number of independent units required to represent the system to accuracy ℓ.

Then it is possible that

$$N_{\text{eff}}(\ell_{\text{coarse}}) = 4$$

while

$$N_{\text{eff}}(\ell_{\text{fine}}) = 5.$$

20 A General Effective-Unit Principle

Proposition 20.1 (Hierarchical Unit Principle). *Suppose N_B physical bodies can be partitioned into K clusters*

$$\mathcal{B} = C_1 \cup \cdots \cup C_K$$

such that each cluster has a well-defined external macrostate and internal corrections smaller than a prescribed tolerance ε. Then the system admits a K-unit effective description to accuracy ε.

Proof. For each cluster C_k, define a macroscopic state Z_k such that the external field produced by the cluster satisfies

$$F_{C_k} = F(Z_k) + R_k,$$

with

$$\|R_k\| \le \varepsilon_k.$$

If

$$\sum_k \varepsilon_k \le \varepsilon,$$

then replacing every cluster by its macrostate changes the total external field by at most ε under the assumed norm and additive error bound. Hence a K-unit effective representation exists to the stated accuracy. □

21 The $N_{\max} \leq 4$ Interpretation

Consider the conditional dimensional allocation rule

$$1 + 2N \leq 9.$$

Then

$$N \leq 4.$$

A naive fifth independent unit would require

$$1 + 2(5) = 11 > 9.$$

However, the physical system contains five bodies while the hierarchical external representation contains four units:

$$\{\mathrm{S}, \mathrm{Me}, \mathrm{V}, \mathrm{EM}\}.$$

Hence the relevant distinction becomes

$$\boxed{N_B = 5 \quad \text{but} \quad N_{\text{eff}} = 4.}$$

This does not prove that celestial mechanics obeys the informational capacity rule. Rather, it provides a concrete physical example of how hierarchical organization can make the number of effective units smaller than the number of microscopic constituents.

22 Bloc Formation

The Earth–Moon subsystem has analogues across several disciplines.

Table 1: Interpretations of the Earth–Moon hierarchical subsystem.

Discipline	Interpretation
Celestial mechanics	Barycentric subsystem with internal relative motion
Physics	Composite system with monopole and higher multipoles
Network science	Strongly coupled local community
Information theory	Hierarchically encoded subsystem
Computer science	Composite object with hidden internal state
Economics	Coalition or aggregated economic unit
Political science	Bloc with internal governance
International relations	Alliance or federation acting externally as one unit
Statistics	Latent-variable aggregation
AI/ML	Hierarchical state representation
Biology	Organism composed of interacting subunits
Philosophy	Whole retaining internal plurality

23 Network Representation

The exact physical graph is the complete graph

$$G_5 = K_5.$$

Thus

$$|V| = 5, \qquad |E| = 10.$$

However, edge strengths differ enormously.
Let

$$w_{ij} = \frac{G m_i m_j}{r_{ij}^2}$$

be a characteristic instantaneous gravitational interaction strength.

A weighted network can therefore display a community structure even though the unweighted graph is complete.

The Earth–Moon edge is structurally distinguished by the bound satellite relationship.

At a coarse level, network contraction gives

$$G_5 \longrightarrow G_{\text{eff}},$$

where

$$V_{\text{eff}} = \{\mathrm{S}, \mathrm{Me}, \mathrm{V}, \mathrm{EM}\}.$$

24 Graph Contraction

Let $C = \{\mathrm{E}, \mathrm{L}\}$.

Contract this cluster into the node EM.

Then

$$K_5/C$$

contains four effective nodes.

The contraction does not assert that Earth and Moon cease to exist. It asserts that their internal edge is moved into the internal structure of the composite node.

This is exactly analogous to the barycentric mechanical transformation.

25 Information-Theoretic Compression

Let the full state be

$$X = (X_{\mathrm{S}}, X_{\mathrm{Me}}, X_{\mathrm{V}}, X_{\mathrm{E}}, X_{\mathrm{L}}).$$

Define the compressed external state

$$Z = (X_{\mathrm{S}}, X_{\mathrm{Me}}, X_{\mathrm{V}}, X_{\mathrm{EM}}),$$

where X_{EM} contains barycentric variables.

The internal state is

$$H_{\mathrm{EM}} = (\boldsymbol{\rho}_{\mathrm{EM}}, \dot{\boldsymbol{\rho}}_{\mathrm{EM}}, \ldots).$$

Hence

$$\boxed{X \longleftrightarrow (Z, H_{\mathrm{EM}})}$$

under an exact coordinate transformation when all required internal variables are retained.

A coarse representation discards or averages some components of H_{EM}:

$$X \longrightarrow Z.$$

The information loss can be measured by conditional entropy

$$H(X \mid Z).$$

An ideal effective representation minimizes

$$H(X_{\text{future}} \mid Z_{\text{present}})$$

rather than necessarily minimizing

$$H(X_{\text{present}} \mid Z_{\text{present}}).$$

Thus predictive sufficiency is more important than lossless compression.

26 Statistical Mechanics of Hierarchy

Let x denote internal microstates of the Earth–Moon subsystem and Z its external macrostate.

Define

$$\Omega(Z) = \{x : \mathcal{C}(x) = Z\}.$$

A coarse-grained entropy is

$$S(Z) = k \log |\Omega(Z)|.$$

This does not imply that orbital mechanics is literally thermodynamic. Rather, it formalizes the distinction between resolved macrovariables and unresolved internal configurations.

The same mathematical logic appears in statistical mechanics, information theory, Bayesian inference, and machine learning.

27 Economics of Composite Units

Suppose Earth and Moon have separate resources

$$W_E, \qquad W_L.$$

If they form a cooperative economic bloc, total resources are

$$W_{\text{EM}} = W_E + W_L + S_{EL},$$

where S_{EL} represents synergies or costs of integration.

If

$$S_{EL} > 0,$$

cooperation generates surplus.

If

$$S_{EL} < 0,$$

coordination costs dominate.

The bloc's external budget constraint may be

$$C_{\text{EM}} + I_{\text{EM}} + D_{\text{EM}} \leq W_{\text{EM}},$$

where consumption, investment, and defensive expenditure are aggregated.

Internally,

$$C_E + C_L = C_{\text{EM}}$$

need not imply equal distribution.

Thus external aggregation does not eliminate internal political economy.

28 Finance and Barycentric Aggregation

Suppose Earth and Moon possess assets with stochastic payoffs X_E and X_L.

The combined payoff is

$$X_{\mathrm{EM}} = X_E + X_L.$$

Its value is

$$V_{\mathrm{EM},t} = \mathbb{E}_t[m_{t,T}(X_E + X_L)].$$

By linearity,

$$V_{\mathrm{EM},t} = V_{E,t} + V_{L,t}$$

under a common stochastic discount factor.

But internal claims can remain distinct.

Thus financial aggregation resembles barycentric aggregation: an external observer can value the combined bloc while internal claims remain separately defined.

29 Welfare Economics

Let Earth and Moon have utilities

$$U_E, \qquad U_L.$$

A cooperative Earth–Moon welfare function is

$$W_{\mathrm{EM}} = \omega_E U_E + \omega_L U_L.$$

The subsystem chooses policies to solve

$$\max \mathbb{E} \int_0^T e^{-\delta t} W_{\mathrm{EM}}(t)\,\mathrm{d}t.$$

The weights

$$\omega_E, \omega_L$$

encode distributive choices.

Thus the fact that the subsystem acts externally as one unit does not solve the internal welfare-allocation problem.

30 Political Science

A composite strategic unit requires an aggregation mechanism.

Let Earth choose policy u_E and Moon choose policy u_L.

External policy is

$$u_{\mathrm{EM}} = \mathcal{G}(u_E, u_L),$$

where $\mathcal{G}$ is a governance rule.

Possible rules include

- unilateral leadership,
- weighted voting,
- bargaining,
- constitutional federation,
- contractual delegation,

- algorithmic coordination.

The mechanical barycenter is uniquely defined by masses.
The political "barycenter" is not.
This is a crucial distinction between celestial mechanics and political aggregation.

31 International Relations and Geopolitics

At the strategic level, the four effective units may be written

$$\mathcal{U} = \{\mathrm{S}, \mathrm{Me}, \mathrm{V}, \mathrm{EM}\}.$$

If these represent technologically capable societies rather than passive astronomical objects, relations may include

$$w_{ij} = (w_{ij}^I, w_{ij}^T, w_{ij}^C, w_{ij}^S),$$

where the components represent information, trade, communication, and strategic relations.
The Earth–Moon subsystem then resembles a federation or alliance whose external policy can differ from its internal distribution of authority.
The physical system therefore supplies a useful mathematical metaphor for hierarchical international organization, while the analogy must not be mistaken for an identity between gravitational and political forces.

32 Warfare Economics and Defensive Allocation

A strategic subsystem facing risk must allocate scarce resources.
Let

$$Y_{\mathrm{EM}} = C_{\mathrm{EM}} + I_{\mathrm{EM}} + D_{\mathrm{EM}} + A_{\mathrm{EM}},$$

where

- C_{EM} is consumption,
- I_{EM} is productive investment,
- D_{EM} is defensive expenditure,
- A_{EM} is astronomical and informational investment.

The opportunity cost of additional security is

$$\frac{\partial C_{\mathrm{EM}}}{\partial D_{\mathrm{EM}}} < 0$$

under a binding resource constraint.
Information can improve defensive efficiency without itself being a physical force.

33 Strategic Conflict

Strategic conflict is represented abstractly as a differential game.
Let the subsystem minimize

$$J_{\mathrm{EM}} = \mathbb{E} \int_0^T \left[L_{\mathrm{risk}}(X_t) + \lambda \| u_{\mathrm{EM}}(t) \|^2 \right] \, \mathrm{d}t.$$

The first term penalizes strategic exposure and the second penalizes costly control.
This formulation is sufficient for studying defensive maneuver, resource competition, and deterrence without specifying operational weapon design.

34 Chemistry and Materials

Any technological strategic interpretation requires material substrates.

Let $\boldsymbol{n}$ denote chemical inventories and let

$$\dot{\boldsymbol{n}} = S\boldsymbol{v}$$

be a stoichiometric reaction system.

Propulsion, energy storage, life support, construction, and radiation shielding depend upon chemical and material constraints.

Therefore the feasible control set is

$$u_i \in \mathcal{U}_i(\boldsymbol{n}, E_i, M_i).$$

No amount of information can produce a physically infeasible maneuver.

This establishes the hierarchy

$$\boxed{\text{Knowledge} \to \text{Decision} \to \text{Feasible Control} \to \text{Physical Motion.}}$$

35 Biology and Hierarchical Organization

Biology provides another example in which many physical constituents form a smaller number of functional units.

Cells form tissues, tissues form organs, and organs form organisms.

Symbolically,

$$\{x_1, \ldots, x_n\} \longrightarrow Z.$$

The macrostate does not imply disappearance of internal components.

Similarly,

$$\{\mathrm{E}, \mathrm{L}\} \longrightarrow \mathrm{EM}$$

at an appropriate external resolution.

The general lesson is that

$$\boxed{\text{microscopic multiplicity} \neq \text{macroscopic unit count.}}$$

36 Psychology and Collective Cognition

If Earth and Moon contain intelligent agents, their collective belief need not equal either constituent belief.

Let

$$\pi_E(X), \qquad \pi_L(X)$$

be separate beliefs.

A collective posterior may be represented schematically as

$$\pi_{\mathrm{EM}} = \mathcal{A}(\pi_E, \pi_L),$$

where $\mathcal{A}$ is an aggregation operator.

For logarithmic opinion pooling,

$$\pi_{\mathrm{EM}}(X) \propto \pi_E(X)^{\omega_E} \pi_L(X)^{\omega_L},$$

with

$$\omega_E + \omega_L = 1.$$

Thus collective cognition is another compression problem.

37 Psychiatric and Cognitive-System Caution

Mathematical models of belief rigidity, instability, disagreement, or memory should not automatically be interpreted as psychiatric diagnoses.

A generic adaptive belief system may be written

$$\pi_{t+1} = (1-\alpha)\pi_t + \alpha\widehat{\pi}_{t+1},$$

where $\widehat{\pi}_{t+1}$ is an evidence-based update.

The parameter α describes responsiveness.

Such equations model information-processing properties. Clinical psychiatric interpretation requires independent medical evidence and should not be inferred from abstract dynamical behavior alone.

38 Philosophy: Parts, Wholes, and Levels of Description

The Earth–Moon construction raises a classical philosophical problem.

Is the Earth–Moon system one thing or two?

The mathematical answer is resolution-dependent.

At the microscopic physical level,

$$N_B = 2$$

within the subsystem.

At a sufficiently coarse external gravitational level,

$$N_{\text{eff}} = 1.$$

Therefore the propositions

“Earth and Moon are two bodies”

and

“the Earth–Moon subsystem is one effective unit”

are not contradictory.

They refer to different levels of description.

This suggests a general principle of ontological hierarchy:

$$\boxed{\text{plurality of parts} \quad \text{can coexist with} \quad \text{unity of effective function.}}$$

39 Cosmological Context

On cosmological scales, write

$$\boldsymbol{r} = a(t)\boldsymbol{x}.$$

The Hubble parameter is

$$H(t) = \frac{\dot{a}}{a}.$$

For a strongly gravitationally bound local system, cosmic expansion is normally negligible compared with local orbital dynamics.

Nevertheless, the conceptual hierarchy extends naturally:

$$\text{Moon} \subset \text{Earth–Moon system} \subset \text{Solar System} \subset \text{Galaxy} \subset \text{cosmological structure.}$$

Effective-unit descriptions therefore depend upon scale throughout astrophysics and cosmology.

40 Statistics and State Estimation

Let the discrete-time state equation be

$$X_{t+1} = f(X_t) + w_t,$$

and observations satisfy

$$Y_t = h(X_t) + v_t.$$

Suppose

$$w_t \sim \mathcal{N}(0, Q_t), \qquad v_t \sim \mathcal{N}(0, R_t).$$

Because gravitational dynamics are nonlinear, possible estimation methods include

- extended Kalman filters,
- unscented Kalman filters,
- particle filters,
- sequential Monte Carlo,
- Markov chain Monte Carlo,
- variational inference.

The hierarchical representation suggests factorizing the posterior:

$$p(X) = p(X_{\text{external}}, X_{\text{internal}}).$$

When justified,

$$p(X) \approx p(X_{\text{external}})p(X_{\text{internal}} \mid X_{\text{external}}).$$

This can substantially simplify inference.

41 Data Science

A practical dataset may contain

$$\mathcal{D} = \{t_k, Y_k, \Sigma_k, A_k\}_{k=1}^n,$$

where A_k identifies the observing apparatus.
The empirical workflow is

$$\boxed{\text{Data} \to \text{Calibration} \to \text{State Estimation} \to \text{Hierarchical Model} \to \text{Prediction} \to \text{Residual Analysis.}}$$

For each approximation order n, define residuals

$$e_k^{(n)} = Y_k - \widehat{Y}_k^{(n)}.$$

A natural loss is

$$L_n = \sum_k (e_k^{(n)})^T \Sigma_k^{-1} e_k^{(n)}.$$

The hierarchy should be judged by predictive improvement, not merely by conceptual elegance.

42 Artificial Intelligence and Machine Learning

The hierarchical decomposition can be learned from data.

Let an encoder map the five-body state to a latent state:

$$Z_t = f_\theta(X_t).$$

A decoder predicts future dynamics:

$$\widehat{X}_{t+\tau} = g_\phi(Z_t).$$

A predictive loss is

$$\mathcal{L}_{\text{pred}} = \mathbb{E}\left[\|X_{t+\tau} - g_\phi(f_\theta(X_t))\|^2\right].$$

To encourage hierarchical compression, add

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \beta \mathcal{L}_{\text{comp}}.$$

A successful model should discover that Earth and Moon share a strongly coupled internal representation while the Earth–Moon barycenter governs much of their external heliocentric motion.

43 Graph Neural Networks

Represent the bodies as graph nodes

$$V = \{\mathrm{S}, \mathrm{Me}, \mathrm{V}, \mathrm{E}, \mathrm{L}\}.$$

Messages are

$$m_{ij} = \phi_\theta(x_i, x_j, r_{ij}).$$

Node updates are

$$x_i' = \psi_\theta\left(x_i, \sum_{j \neq i} m_{ij}\right).$$

A hierarchical graph neural network can first pool

$$\{\mathrm{E}, \mathrm{L}\} \to \mathrm{EM}$$

and then propagate messages among

$$\{\mathrm{S}, \mathrm{Me}, \mathrm{V}, \mathrm{EM}\}.$$

This computational architecture mirrors the physical barycentric decomposition.

44 Hierarchical Reinforcement Learning

For controllable agents, a high-level policy can operate on effective units,

$$u_{\text{high}} = \pi_H(Z),$$

while a low-level Earth–Moon policy handles internal coordination:

$$(u_E, u_L) = \pi_L(X_E, X_L, u_{\text{high}}).$$

Thus

$$\boxed{\text{External Strategy} \to \text{Internal Coordination} \to \text{Physical Controls.}}$$

This is analogous to hierarchical control architectures in robotics and multi-agent systems.

45 Computer Science: State Abstraction

Let the exact state space be $\mathcal{S}$.

Define an abstraction

$$\alpha : \mathcal{S} \to \widetilde{\mathcal{S}}.$$

Two states x_1, x_2 are equivalent if

$$\alpha(x_1) = \alpha(x_2).$$

For prediction, a useful abstraction approximately satisfies

$$p(X_{t+\tau} \mid X_t = x_1) \approx p(X_{t+\tau} \mid X_t = x_2)$$

whenever

$$\alpha(x_1) = \alpha(x_2).$$

Thus the Earth–Moon effective-unit idea is a special case of state abstraction.

46 Algorithm for the Revolutionary Five-Body Problem

```
Algorithm: Hierarchical Revolutionary Five-Body Solver

Input:
    masses of Sun, Mercury, Venus, Earth, Moon
    initial positions and velocities
    observational data
    target tolerance epsilon
    terminal time T

1. Transform Earth and Moon:
       compute Earth-Moon barycenter R_EM
       compute internal coordinate rho_EM

2. Remove total center-of-mass translation.

3. Construct leading Hamiltonian:
       H0 =
           H_K(Sun,Mercury)
         + H_K(Sun,Venus)
         + H_K(Sun,Earth-Moon barycenter)
         + H_K(Earth,Moon)

4. Solve H0 analytically/numerically using Kepler solutions.

5. Construct perturbations:
       planetary interactions
       solar tidal effect on Earth-Moon subsystem
       lunar perturbations
       relativistic corrections
       multipole corrections

6. Average fast variables where justified.

7. Integrate secular dynamics.

8. Reconstruct Earth and Moon from:
       R_EM and rho_EM
```

```
9. Compare predicted observables with data.

10. Compute residual error.

11. If error > epsilon:
       add next perturbative order
       update parameters from observations
       repeat from Step 5

12. Return:
       trajectories
       orbital elements
       posterior uncertainties
       approximation order
       quantified residual error
```

47 Adaptive Error-Control Algorithm

```
Algorithm: Recursive Approximation Control

Set n = 0
Compute X^(0)(t)
Compute error estimate epsilon_0(t)

while max_t epsilon_n(t) > target_tolerance:

    identify dominant omitted interaction

    add interaction to Hamiltonian:
        H^(n+1) = H^(n) + DeltaH_n

    recompute trajectory X^(n+1)(t)

    assimilate new observations

    estimate epsilon_(n+1)(t)

    if error fails to decrease:
        reconsider hierarchy or model assumptions

    n = n + 1

return X^(n)(t), epsilon_n(t)
```

48 A Definition of Solution

For the present problem, insisting upon an elementary closed-form expression for every coordinate is unnecessarily restrictive.

Definition 48.1 (Controlled Hierarchical Solution)**.** *A controlled hierarchical solution of the Revolutionary Five-Body Problem is a sequence*

$$\{X^{(n)}(t)\}_{n=0}^{\infty}$$

together with computable error bounds $\varepsilon_n(t)$ such that

$$\|X(t) - X^{(n)}(t)\| \leq \varepsilon_n(t)$$

over the specified interval and

$$\varepsilon_n(t) \to 0$$

under the convergence assumptions of the approximation scheme.

This notion combines mathematical approximation with empirical falsifiability.

49 Vector Graphic: Hierarchical Solar Architecture

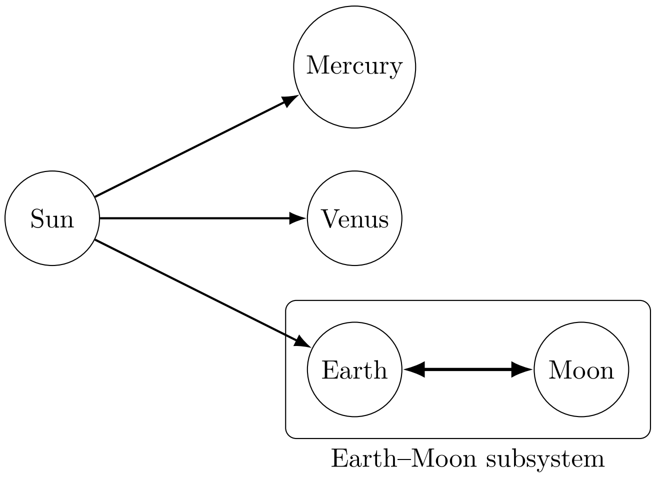

Figure 1: The fifth body enters hierarchically: the Moon is organized into an Earth–Moon subsystem rather than appearing as another independent heliocentric planet.

50 Vector Graphic: Barycentric Decomposition

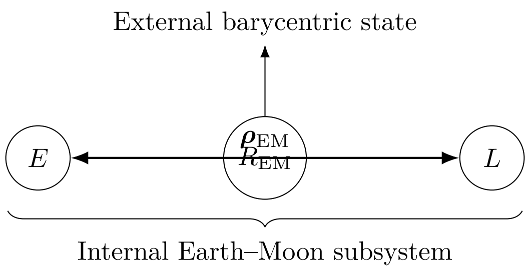

Figure 2: Separation of Earth–Moon external barycentric motion from internal relative motion.

51 Vector Graphic: Five Bodies to Four Effective Units

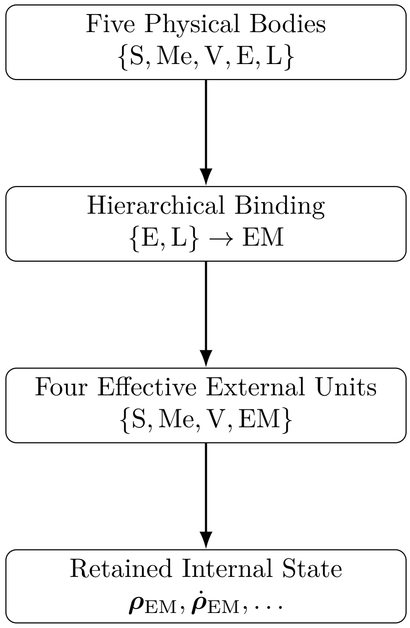

Figure 3: Hierarchical compression changes the effective external unit count without deleting the Moon's internal degrees of freedom.

52 Vector Graphic: Nested Kepler Architecture

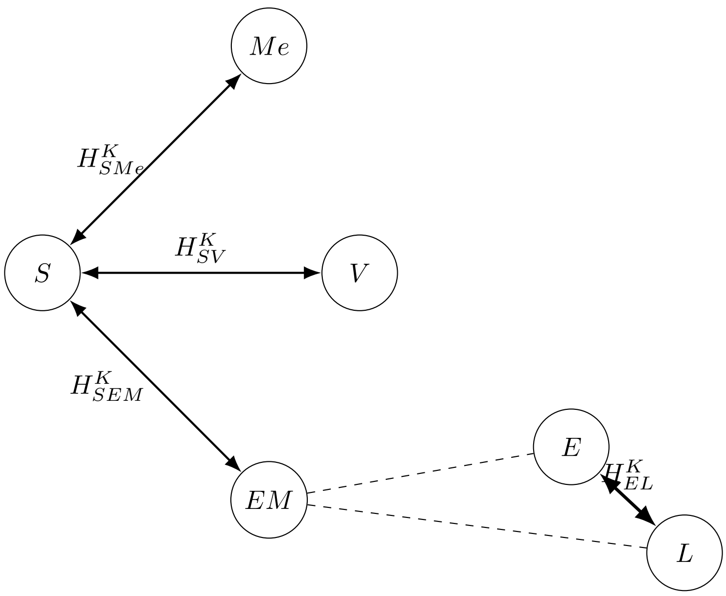

Figure 4: Leading-order decomposition into heliocentric Kepler problems and an internal Earth–Moon Kepler problem.

53 Vector Graphic: Multiscale Dynamics

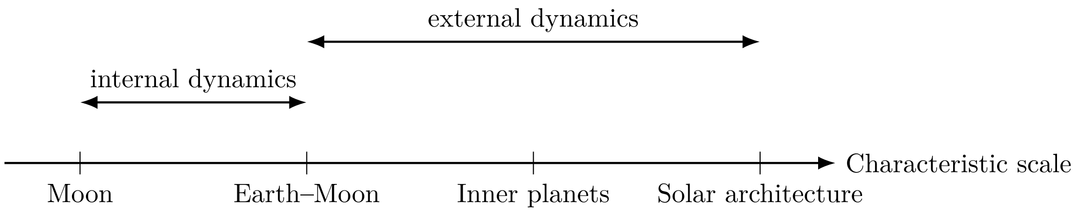

Figure 5: The Revolutionary Five-Body Problem is naturally multiscale.

54 Comparative Table

Table 2: Strategic Four-Body and Revolutionary Five-Body architectures.

Property	Four-body system	Five-body system
Physical bodies	4	5
Position coordinates	12	15
After translation reduction	9	12
Pair interactions	6	10
Satellite subsystem	No	Earth–Moon
External effective units	4	4 at coarse scale
Internal bloc state	None required	Earth–Moon relative state
Natural hierarchy	Solar	Solar + lunar
Time-scale structure	Planetary	Planetary + lunar

55 Generalization to Many Bodies

The same reasoning extends beyond five bodies.

Suppose N_B physical bodies are partitioned into K clusters:

$$\mathcal{B} = C_1 \cup C_2 \cup \cdots \cup C_K.$$

Then

$$N_{\mathrm{eff}} = K$$

at the corresponding resolution.

Necessarily,

$$1 \leq N_{\mathrm{eff}} \leq N_B.$$

A hierarchy of partitions can produce

$$N_B \to N_1 \to N_2 \to \cdots \to N_{\mathrm{macro}}.$$

Thus the central object is not merely body count but a *resolution-dependent hierarchy of effective units.*

56 The Hierarchical Capacity Conjecture

Conjecture 56.1 (Hierarchical Capacity Conjecture)**.** *Suppose a finite-dimensional strategic representation admits at most $N_{\max}$ independent effective units. A physical system containing $N_B > N_{\max}$ bodies can nevertheless admit such a representation if its bodies can be partitioned into no more than $N_{\max}$ dynamically and informationally sufficient composite units.*

For the Revolutionary Five-Body Problem,

$$N_B = 5,$$

while the proposed partition is

$$\{\{\mathrm{S}\}, \{\mathrm{Me}\}, \{\mathrm{V}\}, \{\mathrm{E}, \mathrm{L}\}\}.$$

Therefore

$$N_{\mathrm{eff}} = 4.$$

The conjecture is not established by the Earth–Moon example, but the example provides a concrete system in which it can be investigated.

57 Falsification Criteria

The hierarchical interpretation should be rejected at a specified scale if the neglected internal structure produces errors larger than the required tolerance.

Let

$$X_{\mathrm{full}}(t)$$

be the full five-body trajectory and

$$X_{\mathrm{hier}}(t)$$

the hierarchical approximation.

Define

$$E(T) = \sup_{0 \le t \le T} \|X_{\mathrm{full}}(t) - X_{\mathrm{hier}}(t)\|.$$

The effective-unit approximation is acceptable only if

$$E(T) \le \varepsilon.$$

Thus whether Earth and Moon count as one effective unit is not an absolute statement. It depends upon

$$(T, \varepsilon, \text{observable}).$$

This converts a philosophical question into an empirical and mathematical one.

58 Research Program

A rigorous program for the Revolutionary Five-Body Problem consists of:

1. Construct exact Jacobi/barycentric coordinates for the five-body system.
2. Separate Earth–Moon external and internal motion.
3. Derive the leading Keplerian Hamiltonian H_0.
4. Derive the perturbation Hamiltonian explicitly.
5. Establish the relevant dimensionless small parameters.
6. Apply canonical perturbation and multiple-scale methods.
7. Derive secular equations for orbital elements.
8. Include post-Newtonian corrections where observationally required.
9. Construct error estimates.

10. Assimilate astronomical observations.
11. Compare hierarchical and full numerical integrations.
12. Determine the domains in which $N_{\text{eff}} = 4$ is statistically and dynamically sufficient.
13. Train machine-learning compression models independently and test whether they discover an Earth–Moon latent cluster.
14. Compare the learned hierarchy with the analytically derived barycentric hierarchy.
15. Generalize the theory to additional planets and satellites.

59 Conclusion

The Revolutionary Five-Body Problem begins with

$$\boxed{\{\mathrm{S}, \mathrm{Me}, \mathrm{V}, \mathrm{E}, \mathrm{L}\}.}$$

A generic five-body interpretation treats these as five separate bodies.

The hierarchical interpretation notices that the Moon occupies a special structural position:

$$\mathrm{E} \leftrightarrow \mathrm{L}.$$

The exact barycentric transformation gives

$$(\boldsymbol{r}_{\mathrm{E}}, \boldsymbol{r}_{\mathrm{L}}) \longleftrightarrow (\boldsymbol{R}_{\mathrm{EM}}, \boldsymbol{\rho}_{\mathrm{EM}}).$$

The external Solar-System architecture can consequently be represented at leading order as

$$\boxed{\{\mathrm{S}, \mathrm{Me}, \mathrm{V}, \mathrm{EM}\},}$$

while the internal Earth–Moon state remains separately represented.

Thus

$$\boxed{N_B = 5, \qquad N_{\text{eff}} = 4.}$$

The distinction is fundamental.

Hierarchical compression does not destroy physical degrees of freedom. Instead, it reorganizes them into

$$\boxed{\text{external macrostate} + \text{internal state.}}$$

The leading Hamiltonian becomes

$$\boxed{H_0 = H^K_{\mathrm{SMe}} + H^K_{\mathrm{SV}} + H^K_{\mathrm{SEM}} + H^K_{\mathrm{EL}}.}$$

Corrections are then introduced systematically:

$$H = H_0 + H_{\text{planetary}} + H_{\text{tidal}} + H_{\text{relativistic}} + H_{\text{multipole}} + \cdots .$$

The resulting solution strategy is

$$\boxed{\begin{array}{c}\text{Five physical bodies}\\ \downarrow\\ \text{identify hierarchy}\\ \downarrow\\ \text{Earth–Moon barycentric decomposition}\\ \downarrow\\ \text{four effective external units}\\ \downarrow\\ \text{four leading Kepler structures}\\ \downarrow\\ \text{perturbative corrections}\\ \downarrow\\ \text{astronomical observations}\\ \downarrow\\ \text{Bayesian inference}\\ \downarrow\\ \text{error-controlled prediction.}\end{array}}$$

The conceptual result extends beyond celestial mechanics:

$$\boxed{\text{number of physical constituents} \neq \text{number of effective units.}}$$

A system may contain more microscopic bodies than its macroscopic representation contains independent units.

The Earth–Moon pair supplies a particularly transparent astronomical example because the distinction is supported simultaneously by barycentric mechanics, scale separation, multipole expansion, network contraction, and hierarchical state representation.

The Revolutionary Five-Body Problem therefore does not merely add one body to the Strategic Four-Body Problem.

It changes the question.

Instead of asking only

> How do five bodies move?

we ask

> Under what physical, informational, and strategic conditions can five bodies organize into four effective units, while preserving the internal dynamics necessary to reconstruct and predict the complete system?

That question provides a bridge from few-body celestial mechanics to a general theory of hierarchical organization.

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Glossary

Barycenter
The mass-weighted center of a system of bodies.

Barycentric Compression
Representation of a subsystem's external motion through its barycenter while its internal relative coordinates are modeled separately.

Celestial Hierarchy
A nested gravitational architecture in which smaller bound systems participate in larger orbital systems.

Composite Unit
A collection of physical bodies represented as one external effective unit at a specified scale.

Controlled Hierarchical Solution
A sequence of increasingly accurate hierarchical approximations accompanied by explicit error estimates.

Earth–Moon Barycenter
The mass-weighted center of the Earth–Moon subsystem,

$$\boldsymbol{R}_{\mathrm{EM}} = \frac{m_{\mathrm{E}}\boldsymbol{r}_{\mathrm{E}} + m_{\mathrm{L}}\boldsymbol{r}_{\mathrm{L}}}{m_{\mathrm{E}} + m_{\mathrm{L}}}.$$

Effective Strategic Unit
A body or subsystem that can be represented by a common macroscopic state for a specified strategic problem.

External State
Variables describing the behavior of a subsystem relative to the larger system.

Graph Contraction
Replacement of a collection of graph nodes by a composite node while retaining an effective representation of external connections.

Hierarchical Binding
Organization of multiple bodies into a subsystem with strong internal structure and a simpler external representation.

Hierarchical Capacity Conjecture
The conjecture that a system with more physical bodies than an effective-unit capacity can remain representable when its bodies form sufficiently few composite units.

Hierarchical Compression
Reduction of effective external complexity by representing strongly coupled subsystems through macroscopic states while retaining relevant internal variables.

Internal State
Variables describing dynamics inside a composite subsystem.

Keplerian Structure
A leading-order two-body gravitational subsystem described by the Kepler problem.

Multipole Expansion
Representation of the external field of a spatially extended system as a monopole plus progressively smaller higher-order corrections.

Physical Body Count N_B
The actual number of separately identified physical bodies.

Effective Unit Count N_{eff}
The number of independent units required by a specified coarse-grained representation.

Resolution-Dependent Unit Count
The effective number of units required at a particular spatial, temporal, observational, or predictive resolution.

Revolutionary Five-Body Problem
The hierarchical Sun–Mercury–Venus–Earth–Moon problem developed in this paper.

Secular Dynamics
Slow cumulative orbital evolution obtained after averaging over faster orbital phases.

Strategic Astronomy
An approach in which astronomical observation is treated as a recursive information-generating process connected to inference and decision.

Strategic Four-Body Problem
The predecessor framework involving four mutually interacting and potentially information-processing bodies.

Tidal Interaction
Differential gravitational forcing across a spatially extended subsystem.

The End